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WHICH TIES MATTER WHEN? THE CONTINGENT EFFECTS OF INTERORGANIZATIONAL PARTNERSHIPS ON IPO SUCCESS

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This paper investigates the contingent value of interorganizational relationships at the time of a young firm's initial public offering (IPO). We compare the signaling value to young firms of having ties with two types of interorganizational partnerships: endorsement relationships such as those with venture capital firms and investment banks, and strategic alliance partnerships. We propose that, under different equity market conditions, potential investors in an issuing firm attend to different types of uncertainty; attention to these different types of uncertainty affects investors' perceptions of the relative value of a young firm's different kinds of endorsements and partnerships and, hence, IPO success. Results from a sample of young biotechnology firms show that ties to prominent venture capital firms are particularly beneficial to IPO success during cold markets, while ties to prominent investment banks are particularly beneficial to IPO success during hot markets; a firm's strategic alliances with major pharmaceutical/health care firms did not have such contingent effects. Implications for understanding the contingent value of interorganizational ties are discussed. Copyright © 2003 John Wiley & Sons, Ltd.

In recent years, there has been growing interest in understanding how the social context in which firms are embedded affects their behavior and performance. Building on the general axiom that economic action does not occur in isolation (Granovetter, 1985; Laumann, Galaskiewicz, and Marsden, 1978; Galaskiewicz and Marsden, 1978), studies have generally touted the benefits of embeddedness. There is a growing body of research suggesting that embeddedness does not necessarily lead to positive outcomes; rather, the value of ties may vary depending upon the situation. While organizational scholars have begun to examine the contingent value of ties for interpersonal networks (e.g., Burt, 1992; Podolny and

Baron, 1997), there is less understanding of this possibility for interorganizational networks (Gulati and Westphal, 1999).

Understanding whether, how, and what circumstances modify the effects of embeddedness in various kinds of networks for young firms is an important undertaking, considering recent scholarship that has indicated that ties can have deleterious effects on firm outcomes (e.g., Gargiulo and Benassi, 1999). Attempting to understand the contingent value of embeddedness should help scholars and those involved in entrepreneurial ventures better understand how to negotiate varying market conditions. This also enriches the network perspective by specifying the conditions under which social networks affect organizational performance.

We investigate the research question, which ties matter when, in the context of young firms undergoing their initial public offering (IPO). Prior research in finance and strategy has

Key words: entrepreneurship; networks; strategic alliances; venture capital firms; investment banks; IPO performance

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shown that ties to prominent firms enhance new venture performance (Carter and Manaster, 1990; Baum, 1996). Further, organizational scholars have suggested that ties to prominent organizations can signal a firm's quality to key external resource holders, which affects IPO performance (Stuart, Hoang, and Hybels, 1999). However, prior research has not examined if and when the effects of interorganizational embeddedness vary for young firms, and so yield different consequences for new venture performance.

In this paper, we propose that network ties are not uniform in their effects on firm outcomes, such as IPO success, but rather vary across different types of network ties a firm has developed as well as the uncertainty associated with the relevant equity market. Together, these two dimensions, network tie type and market uncertainty, form the basis of our exploration into the contingent value of interorganizational relationships for entrepreneurial firms. We focus on three types of network ties: ties to prominent venture capital firms (VCs), ties to prominent investment banks, and ties to prominent strategic alliances. The first two are what we call 'endorsement' ties and the last are strategic alliances. We focus on an exogenous form of market uncertainty as a moderator of network effects that has been underexplored in prior strategy research: uncertainty associated with the favorability of the equity markets. Our key claim is that the benefits of each type of firm tie not only vary across each other but will be moderated by market uncertainty such that certain types of ties will matter more or less under certain conditions. The theoretical underpinning for our claim is the proposition that different types of market uncertainty focus investor attention on different sets of factors. Since network ties provide important signals of a firm's potential, they become more or less important, depending upon the investor concerns in the specific market context.

The present research contributes to research on organizational strategy in several respects. First, our work builds upon prior research which has emphasized the signaling value to young firms of having prominent affiliations (e.g., Podolny, 1994; Podolny, Stuart, and Hannan, 1996; Stuart *et al.*, 1999). Much of the research on interorganizational relationships has revolved around the general axiom that more embeddedness is 'better'—that having ties to powerful and central

actors is particularly beneficial. The present study challenges this basic assumption by examining the conditions under which more is indeed better for entrepreneurial firms and which kinds of specific ties are more beneficial than others. Our theory suggests that, depending upon the market conditions in which a young firm performs (here, goes public), certain types of prominent ties may be more or less beneficial to IPO performance. Thus, we contribute to research on social networks by examining the contingent value of interorganizational embeddedness.

Second, we contribute to research on organizational strategy that has proposed an attention-based view of firm behavior (Simon, 1947/1997; Ocasio, 1997). Prior research has suggested that certain prevailing guidelines or 'logics' associated with an industry differentially focus executive attention and decision-making (Ocasio, 1997; Thornton and Ocasio, 1999; Thornton, 2001). We build upon these ideas and suggest that, depending upon the favorability of the equity markets, different logics may prevail, affecting investor decision-making and hence IPO performance. We propose that investors are generally concerned about making two types of errors: investing in unworthy firms and missing good opportunities. We propose that the extent to which investors attend to one or the other of these concerns is likely to vary with how favorable or unfavorable the market is for new issues. We argue that, depending upon which concern prevails, different types of interorganizational ties will have different levels of signaling value, affecting investor decisions and hence IPO success.

Third, this study enhances our understanding of how entrepreneurial firms secure material resources that are critical to firm survival. Although prior research by strategy and organizational scholars has centered on the value to firms of securing resources through interorganizational partnerships over a firm's life course (e.g., Baum and Oliver, 1991), rarely have scholars considered how this central activity of securing resources may be largely influenced by factors associated with the external marketplace. Understanding how key stakeholders, such as investors, respond to changes associated with the public markets and undertake resource allocation decisions that affect the performance of young firms are questions that are central to research on strategy and entrepreneurship.

THEORY AND APPLICATION TO THE EQUITY MARKETS

In recent years, there has been a burgeoning literature that has suggested the significance of various kinds of interorganizational ties, including ties to prominent organizations. One benefit to young firms in developing partnerships with well-established firms is access to valuable information and capabilities that can enable young firms to overcome liabilities of newness and/or smallness (Baum and Silverman, 1999). Additionally, networks can be beneficial to firms by signaling the quality of a new venture by mitigating uncertainty in the eyes of third parties (Stuart *et al.*, 1999). Extending this central idea that ties to prominent actors mitigate uncertainty, we propose that the value of a young firm's ties should vary not only across various types of ties but also depend upon the different types of uncertainty a firm faces. As we will suggest, different types of uncertainty may raise different types of concerns for key outsiders, such as investors, affecting the value of a firm's partnerships.

For outside resource-holders such as public investors, uncertainty associated with a firm can arise due to characteristics associated with the firm itself, such as firm age or location (e.g., Sorenson and Audia, 2000). In addition, uncertainty can arise from *exogenous* sources such as natural events, changes in preferences, or regulatory changes (Sutcliffe and Zaheer, 1998). How favorable or unfavorable the market is for equity offerings is one critical aspect of uncertainty for investors. As research in the biotechnology industry, the context of the present study, has shown, the receptivity of the equity markets for biotechnology offerings has ebbed and flowed over the years, affecting the preferred timing of IPOs (Lerner, 1994). When the market window is relatively open (i.e., 'hot') for equity offerings, the potential upside for both firms and investors is much greater than when the market window is relatively closed (i.e., 'cold'). Investors' decisions to provide financial resources to a young firm is thus nested in uncertainty associated with the receptivity of the equity markets—what we will call 'equity market uncertainty.'

We propose that equity market uncertainty entails two overarching types of concerns for investors: investing in 'bad' (or low-potential) firms or missing 'good' (or high-potential) opportunities. Although both of these concerns

surround any investment decision, we propose that investors' attention shifts more toward one or the other, depending upon the receptivity of the equity markets. When the equity markets are relatively hot for new issues, many firms try to go public, making the probability of, and hence concern about, investing in unworthy firms more salient. When the equity markets are relatively cold for new issues, few firms try to go public, making the probability of, and hence concern about, overlooking good firms more salient. Indeed, prior research has shown that when the equity markets are relatively hot, investors are overly optimistic about the potential of young firms (Ritter, 1984). At such times, investors' concerns revolve around 'gullibility' or investing in firms when they should not (i.e., making a Type II error). On the other hand, when the equity markets are relatively cold and few firms try to go public, investors' concerns revolve around 'blindness' or overlooking firms which they should invest in (i.e., making a Type I error) (cf. Rosenthal and Rosnow, 1991; Sah and Stiglitz, 1986, 1988).

This perspective on investor decision-making is consistent with an attention-based view of the firm. As Ocasio (1997: 189) describes, attention encompasses 'the noticing, encoding, interpreting and focusing of time and effort by organizational decision-makers.' In the present context, whether and how much investors decide to invest depends upon the types of concerns they attend to which may be influenced by the particular context (e.g., market context) they find themselves in (Ocasio, 1997). Extending these ideas, we suggest that investors can resolve specific concerns (such as investing in bad firms) by attending to information associated with the ties of the focal firm. Here, we consider two types of ties: endorsement relationships such as those with VCs and investment banks and strategic alliance partnerships. We propose that each type of tie mitigates different types of uncertainty and thus varies in importance at different times.

Endorsement relationships

Endorsement relationships are interorganizational relationships a firm has with another organization that serves as an intermediary between the focal firm (here, the issuing firm) and a third party

(here, public investors). Endorsement by powerful organizations can enable young firms to overcome the external liability of newness problems that stem from their lack of a favorable reputation (Rao, 1994; Rao and Drazin, 2000; Thornton, 1999). Endorsement is a particular concern for firms that face mediated markets, such as the initial public offering, which are characterized by significant ambiguity regarding firm valuation and which depend on highly visible critics, such as investment analysts, who wield tremendous influence over investor behavior. As recent research has found, firms that fail to gain reviews by critics who specialize in the firm's product can convey illegitimacy to the marketplace, affecting firm outcomes (Zuckerman, 1999).

In the present IPO context, endorsement relations include ties with VCs, accounting firms, law firms, and investment banks (Bochner and Priest, 1993). Among these many types of endorsing organizations involved in a firm's IPO, studies in finance and organizational strategy have demonstrated that two types of organizations play a particularly significant role: VCs and investment banks (Carter and Manaster, 1990; Jain and Kini, 1995; Stuart *et al.*, 1999). And, while prior research has established that ties to prominent organizations such as these can enhance IPO success in general, we suggest that the value of these ties will be moderated in important and systematic ways by equity market uncertainty. In particular, we expect that the signaling value of a tie to a particular partner will be greatest when that partner is most carefully attending to the IPO market. Since different types of partners attend closely to the equity market at different times, we expect to find variance associated with the value of an IPO firm's ties.

VC endorsement

Prior research has demonstrated that VC backing increases the likelihood that a firm will have a successful IPO. For example, studies demonstrate that VC partnerships provide financial resources and expertise that serve as important signals of new venture quality which positively impacts IPO success (Megginson and Weiss, 1991). Studies have also demonstrated that VC quality is associated with IPO performance: the greater the amount of monitoring by a VC, the lower the underpricing—the spread between issuing price and offering

price shortly after public trading begins—which is one measure of IPO performance (Lin, 1996; Jain and Kini, 1994).

In addition to helping certify the present value of the issuing firm, endorsement by a VC signals the firm's *future* value. Empirical studies show that the presence of VCs improves a young company's chances of survival in the post-IPO period (Khurshed, 2000). The logic is that in addition to providing financial experience to a young firm in the form of knowledge regarding incentive and compensation systems and deal structuring, VCs closely monitor their companies *following* their initial investment (Sahlman, 1990; Gorman and Sahlman, 1989). Therefore, the activities in which VCs are expected to engage in the future, such as recruiting senior managers and developing business strategy (Bygrave and Timmons, 1992; Hellman, 1998; Hellmann and Puri, 2000) should affect post-IPO performance as well. In sum, a firm's partnership with a reputable VC should signal both the present and future quality of a young firm.

The research summarized above on the effect of VC endorsement on IPO success assumes that the VC effect on firm IPOs is uniform at all times. This may not always be the case: market upturns and downturns may alter this effect. As described, partnerships with high-quality VCs signal to the investment community that the new venture is a 'good bet' both at the time of IPO and in the future. Since VC ties alleviate uncertainty regarding the decision to invest in an IPO firm, it is also likely that the signals such ties provide will vary depending upon the types of uncertainty surrounding the investment decision.

Recent research on the decision-making processes of VCs lends insight into how the signals associated with VC ties may differ depending upon the uncertainty that characterizes the equity markets. During hot markets, 'too much money will chase too few deals' (*The Economist*, 1997). As this quotation by Bill Hambrecht, chairman of Hambrecht and Quist, a San Francisco VC, suggests, VCs are likely to be overly optimistic about the upside of firms going public when the markets are favorable. And, with so many firms trying to go public during a hot market, the amount of information that a VC must process to make the decision to take a firm public is likely greater than when fewer firms attempt to go public. Indeed, extensive research on VC decision-making has shown that it is precisely these two conditions—overconfidence

and information overload—that can undermine the accuracy of VC decision-making (Zacharakis and Shepherd, 2001; Zacharakis and Meyer, 1998).

These differences in how and when VCs carefully and accurately attend to the IPO market may affect investor perceptions of the value of VC partnerships. During a cold market, when VCs are not overwhelmed by a hot market's 'fools rush in' phenomenon, investors are likely to attribute greater value associated with a focal firm's tie to a prominent VC. A cold market environment stands in contrast to the information-laden and hyper-competitive environment that can lead to lower attention spans and suboptimal VC decision-making during hot markets (for reviews of research on VC decision-making, see Zacharakis and Meyer, 2000; Fried and Hisrich, 1994). The time and attention taken in the evaluation process during cold markets should result in fewer missed opportunities, or Type I errors, made by VCs. Therefore, while the evaluation expertise of VC endorsements may, in general, signal firm quality to investors, we expect investors to perceive greater value associated with such ties during cold markets.

Hypothesis 1: Endorsement by a prestigious VC should be particularly beneficial to the success of a young company's IPO when the equity markets are cold.

Underwriter endorsement

Prior studies have established a clear role for underwriter reputation in IPOs. Both finance and organizational scholars have demonstrated that firms with prestigious underwriters are more likely to have successful IPOs (Carter and Manaster, 1990; Stuart *et al.*, 1999). Additionally, finance scholars have demonstrated that high-prestige investment banks are unlikely to undertake speculative issues due to the legal liabilities and potential loss of reputational capital associated with such deals (Beatty and Ritter, 1986; Carter and Manaster, 1990; Tinic, 1988). Given this risk profile, prestigious investment banks generally prefer the seasoned equity market as their 'natural playground' over the IPO market (Wolfe, Cooperman, and Ferris, 1994; Bae, Klein, and Bowyer, 1999). And, when prestigious investment banks do engage in the IPO market, they tend to underwrite low-risk IPOs, compared to high-risk

IPOs (Tinic, 1988; Hayes, 1971), with the hope that such relationships will lead to opportunities for larger and more lucrative deals in the future.

These findings from the finance literature all point to the tendency for prestigious investment banks to switch into and out of engaging in the IPO market, depending upon the receptivity of the equity market. Unlike prestigious VCs who specialize in a limited number of industries and engage often in new issues (Jain and Kini, 1995), prestigious investment banks offer a wider range of financial instruments and so have greater choice as to what types of deals they engage in (List, Platt, and Rombel, 2000). Further, unlike prestigious VCs, who take firms public that they have previously invested in, prestigious investment banks tend to be quite selective in picking firms at the IPO stage since it is the *post*-IPO deals that will generate the most attractive returns for them. And, since prestigious investment banks are loath to incur significant risk, they are more likely to engage in doing deals when the equity market is hot than when it is cold for new issues.

These differences in when and how extensively prestigious investment banks attend to the IPO market may affect the types of signals associated with their endorsement. Since prestigious investment banks attend most closely to the equity markets when the market is hot, a young firm's endorsement by a prestigious underwriter should be particularly helpful in alleviating investors' concerns regarding the possibility of investing in bad deals, or making a Type II error. Public investors, like investment banks, face the challenge of choosing between many possible firms to invest in when the equity markets are favorable since the pool of firms trying to go public is relatively large (Ritter, 1984). And, as is the case with investment banks, the number of firms that public investors can support is limited due to their own financial constraints (Puri, 1999). During hot markets, then, considering the large number of firms seeking financial capital and the limited number of firms that prestigious investment banks can actually underwrite, a young firm's partnership with a prestigious underwriter should signal to potential investors that the issuing firm is a relatively good (i.e., low-risk) bet, reducing investors' concerns regarding Type II errors. Indeed, since prestigious investment banks are extremely avoidant of risky ventures (Beatty and Ritter, 1986; Carter and Manaster, 1990) and prefer to engage in post-offering deals (Wolfe *et al.*,

1994), investors may infer that the firm a prestigious underwriter chooses to endorse during a hot market has long-term potential—in short, that it is a worthwhile investment.

Hypothesis 2: Endorsement by a prestigious underwriter should be particularly beneficial to the success of a young company's IPO when the equity markets are hot.

Strategic alliances

A substantial stream of strategy research has examined the value to firms of having strategic alliances. Strategic alliances may provide opportunities for codevelopment or the sharing of capital, technology, or firm-specific assets (Gulati, 1999). Much of this research has focused on the performance implications of strategic partnerships in the context of established firms (e.g., Mowery, Oxley, and Silverman, 1996). Some recent work, however, has explored the value of strategic alliances for entrepreneurial firms (e.g., Baum, Calabrese, and Silverman, 2000; DeCarolis and Deeds, 1999). For young firms in the biotechnology industry, having strategic alliances with prominent pharmaceutical and health care organizations in particular can send powerful signals to outsiders (Stuart *et al.*, 1999).

One of the biggest challenges for young biotechnology companies is to advance products through a long product development cycle to generate revenues (Powell, Koput, and Smith-Doerr, 1996). This is a major concern for investors as well. In the biotechnology industry, it takes 7–10 years for a firm to advance from basic R&D through clinical trials and the FDA approval process (Deeds, DeCarolis, and Coombs, 1997). Due to these long product development cycles and the fact that the needs for cash are in the hundreds of millions of dollars, biotechnology firms tend to be far from generating revenues when they try to go public (Pisano, 1991). One way that biotechnology firms have tried to alleviate investor concerns regarding the viability of their products is to ally with major pharmaceutical and health care companies that engage in downstream activities and so have significant marketing and sales expertise as well as financial capital that can support the firm through this lengthy process (Pisano, 1990, 1991).

Since these types of alliances serve the important function of mitigating uncertainty for resource-holders, we propose that the signal associated with

a firm's strategic alliances may vary with the types of uncertainty that characterize different market situations and the extent to which these alliance partners are most actively engaged in evaluating young firms. During hot markets, a young biotechnology firm's concerns regarding its ability to sustain the long discovery and development process are not as acute as during cold markets when the availability of funding is much more scarce. During such times, young biotechnology firms turn to other options for funding, such as interfirm collaborations, to sustain their efforts (Lerner and Merges, 1996). Therefore, during cold markets, prominent pharmaceutical/health care firms necessarily attend more closely to the potential of young biotechnology firms since so many firms come knocking on their doors for resources (Lerner and Merges, 1996). These concerns about securing financial resources when the equity markets are unfavorable are likely transparent to investors.

This difference in how and when alliance partners carefully evaluate new biotechnology ventures should affect the strength of the signal that is associated with their tie. During cold markets, the signaling value of the strategic alliance is stronger since these well-established firms have paid particularly careful attention to the potential of many young biotechnology firms so as to avoid missing a worthwhile investment (Pisano and Mang, 1993)—which is precisely the same type of Type I concern that plagues investor decision-making during cold markets. Indeed, the fact that a major strategic alliance partner *chose* a particular firm, among the many young firms that were searching for funding during an unfavorable market, may send a particularly powerful signal to outsiders, such as investors, that the young biotech firm is a good bet—that it is a firm worth investing in. Therefore, while, in general, a tie to a major pharmaceutical/health care organization may signal firm quality to outsiders, this should be particularly true when the signal-maker's attention is focused intently on evaluating the potential of young firms, as is the case during cold markets.

Hypothesis 3: Strategic alliances will be especially beneficial to the success of a young company's IPO when the markets are relatively cold for equity offerings.

In summary, when examining the idea that interorganizational ties have differential effects

		Network Factors: Tie Type		
		Endorsement Relations		Strategic Alliances
		Venture Capital Firms	Lead Investment Bank	Downstream Alliances
Contextual Factors: Equity Market Conditions	Hot	H1: Less Positive	H2: More Positive	H3: Less Positive
	Cold	H1: More Positive	H2: Less Positive	H3: More Positive

Figure 1. A contingency perspective on the effects of interorganizational partnerships on IPO success

on IPO performance, we consider two questions: which ties matter (network factors) and when ties matter (contextual factors). In this study of firms undergoing IPOs, the contextual dimension was operationalized as the extent to which the equity markets are favorable to new issues—in simplified terms, whether the equity markets are ‘hot’ or ‘cold’ for new issues. The network dimension has been broken down into the following types of ties: VC partnerships, ties with an underwriter, and strategic alliances. Figure 1 depicts this two (hot vs. cold equity market) by three (tie type) contingency framework and summarizes our predictions.

METHOD

Sample and data collection

Our sample frame includes U.S. biotechnology firms that were founded between 1961 and 1994. Of these 858 firms, 299 went public between 1979 and 1996. Approximately 86 percent of the public firms specialized in the development of therapeutics and/or human diagnostics; the majority of the remaining firms specialized in agriculture and/or other biological products, generally with the intention of engaging in therapeutic applications in the future. The average time to IPO was 4.87 years.

We compiled our data from both published and unpublished sources, striving to be as thorough as possible, yet focused on true, dedicated biotechnology firms. Our primary list of public biotechnology firms was obtained from the *BioWorld Stock Report for Public Biotechnology Companies* in 1996 ($n = 281$). Unlike other sources (e.g.,

BioScan), this listing does not include large corporations (e.g., General Electric) that participate tangentially in the biotechnology industry; hence, ours is a narrower definition of biotechnology than that employed by other researchers (e.g., Barley, Freeman, and Hybels, 1992) and is in line with more recent research on the industry (e.g., Powell *et al.*, 1996). Further, to guard against sample selection bias associated with this listing, we collected information on firms that went public in the same time frame as our sample but that did not survive in their original form by 1996. To do this, we obtained information from organizations that specialize in conducting research on the biotechnology industry, including BIO, the North Carolina Center for Biotechnology Information, Recombinant Capital (ReCap), and the Institute for Biotechnology Information. We also compared three editions of *Biotechnology Guide USA* (Dibner, 1988, 1991, 1995). From these sources, we identified an additional 18 dedicated U.S. biotechnology firms that went public but were not in existence in their original form in 1996. Such firms had experienced name changes, merged, or had been acquired. These firms were founded in the same time period and all had gone public by the end of 1996.

We also collected information on biotechnology firms that were founded in the same time period as our sample but that did not go public by 1996 ($n = 468$) from the 1998 edition of the Institute for Biotechnology Information (IBI) database. We added to this list private biotechnology companies that were listed as ‘dead,’ merged, or acquired in the first three editions of the *Biotechnology Guide*

USA (Dibner, 1988, 1991, 1995) and that had a founding date in the same time period as our core sample ($n = 91$). Combining these private firms with our sample of firms that did go public yielded a final combined sample size of 858 firms.

Dependent measures

IPO success measures

Our measure of IPO success was calculated based upon four different financial measures.¹ First, we obtained the value for a firm's *net proceeds* from the first page of the final prospectus. This is the amount of cash the firm receives as a result of the offering, less costs incurred during the IPO process. Second, we calculated the *pre-money market valuation* of our firms, which is an indicator of IPO success employed in previous organizational and strategic management research (Stuart *et al.*, 1999). The pre-money market value is calculated as follows:

$$V^* = (p_u q_t - p_u q_i)$$

where p_u is the final IPO subscription price as indicated on the firm's final prospectus, q_t measures the number of shares outstanding, and q_i is the number of shares offered in the IPO. This is the firm's market valuation less the proceeds to the firm as a result of the IPO. V^* is therefore the market valuation of the biotechnology firm just preceding the first day of trading. Third and fourth, we calculated each firm's *90-day market valuation* and *180-day market valuation* after the IPO to gauge the early success of the firm's offering. We used the same formula for pre-money market valuation but substituted the post-IPO price at 90 days out and then at 180 days out for p_u in the formula. Since these four financial measures were highly correlated with one another (Cronbach's $\alpha > 0.88$), these measures were standardized and the mean was taken to create one financial indicator of IPO success.

Independent measures

Equity market uncertainty

We employed a financial index developed by Lerner (1994) and widely cited and used in finance

and strategy literature (e.g., Baum *et al.*, 2000; Stuart *et al.*, 1999; Zucker, Darby, and Brewer, 1994), which gauges the receptivity of the equity markets to biotechnology offerings. Lerner's (1994) index was constructed based on an equal amount of dollar shares of 13 publicly traded, dedicated biotechnology firms. The findings of Lerner's (1994) study imply that an industry-specific index is the preferred method of capturing the favorability of the equity markets, as times of high valuations vary across industries and not always in complete conjunction with the general market. The index included in our models is a finer-grained measure of market conditions than has been used recently in research on IPOs in the biotechnology industry (in which dummy variables for 'cold' market years were included) (e.g., DeCarolus and Deeds, 1999). We used the value of Lerner's *equity index* at the end of the month prior to the IPO date for each of our firms as our indicator of industry uncertainty at the time of the IPO. The equity market measure thus ranges from low to high or from cold to hot (unfavorable to favorable).

VC partnerships

VC prominence was measured by an indicator of whether a firm's VC partnerships were prominent. To do this, we created lists of prominent VCs for each IPO year in our dataset as follows: we obtained rankings of VCs from VentureXpert, a Securities Data Corporation database; rankings were based on total dollars invested by each VC in each of the 18 years that comprise the timeframe of our dataset. Firms were coded as 1 if any of the biotechnology firm's VCs with a minimum of a 5 percent stake were listed as among the top 30 on the list of prominent VCs for the year *prior* to the firm's IPO date and 0 otherwise.

Lead underwriter

Underwriter prestige was measured using an index developed by Carter and Manaster (1990) and then updated by Carter, Dark, and Singh (1998). The measures are based on analyses of investment banks' positions in the 'tombstone' announcements for IPOs and have been cited widely in both finance and organizational and strategic management research (e.g., Bae *et al.*, 1999; Podolny, 1994; Rau, 2000; Stuart *et al.*, 1999); this information was available for all but 25 of the underwriters

¹ We thank an anonymous reviewer for the suggestion to construct a consolidated financial measure of IPO performance.

in our dataset (accounting for 55 of our firms), yielding rankings for 244 firms. Mann–Whitney and Kolmogorov–Smirnov tests indicated that the firms for which this information was not available did not differ significantly from those which had this information along any of our main independent variables of interest. Carter and colleagues' indexes were created by looking at the hierarchy of investment banks as presented in the 'tombstone announcements' for IPOs that appear in *Investment Dealer's Digest* or *The Wall Street Journal*. The authors assigned the highest integer rank (9) to the first-listed underwriter, the second highest integer rank (8) to the next-listed underwriter(s), and so on. Taking the second tombstone announcement, they checked to see if any underwriter not listed on the first one was listed above any underwriter that *was* listed on the first one. If this was the case, the new, more highly ranked underwriter was assigned the rank of the superseded underwriter, and the superseded underwriter and all lower-ranked underwriters were shifted one point down on the scale. This continued until all IPOs were exhausted. When more than 10 categories became necessary to preserve the hierarchy presented on the tombstones, decimal increments were employed. Eventually, the scale as presented in Carter *et al.* (1998) is incremented in units of 0.125. Scores may assume a value ranging from 0, indicating lowest prestige, to 9, indicating highest prestige. In our dataset, the mean score was 7.63. Carter and Dark's (1992) analyses suggest that these measures provide a finer-grained evaluation than a simpler market share alternative (e.g., Megginson and Weiss, 1991). The name of the lead investment bank came from the front page of the final prospectuses.

Strategic alliances

We calculated the total number of alliances the firm had to prominent pharmaceutical/health care organizations. To determine which institutions were prominent, we generated 18 lists, one for each IPO year, of the top pharmaceutical and healthcare organizations by sales since 1979, using COMPUSTAT. International companies are only ranked by COMPUSTAT from 1988 on, so our rankings are based on the top 30 U.S. organizations from 1979 to 1987 and from the top 30 U.S. and international organizations from 1988 to 1996. For each year, we coded the top 30 organizations in that

given year as prominent. We supplemented our lists with major pharmaceutical and health care companies that were private or based in Europe or Japan that were not listed in COMPUSTAT but were listed in PharmaBusiness and had comparable sales, since many young biotechnology firms rely on international resources for support and talent. The number of prominent *strategic alliances* was measured as the number of alliances with prominent pharmaceutical and/or health care companies the year prior to the offering, as defined above.

Control variables

We included controls for *firm size* and *firm age*, consistent with prior research on IPOs. We measured *firm age* as the number of years since the firm was founded and *firm size* as the total number of employees as listed in the final prospectuses. And, while not a direct indication of the size of the firm, the amount of *private financing* the firm received prior to the IPO does provide a reliable measure of the success the firm has had in the past in securing financial capital and so is an indicator of the firm's potential for growth as well. Our measure of *private financing* was calculated by adding up the rounds of financing listed in the final prospectuses. This measure was adjusted to constant 1996 dollars and logged in our analyses.

Further, we coded our firms for their geographical *location*. Young firms located in areas that are rich with industry-related activity will likely have greater access to resources, including qualified personnel, suitable lab space, and technology, that can give them an advantage over other young firms (Saxenian, 1994). Given the research and technology centers of the United States, locational advantage is likely to accrue to firms that choose to operate in central areas like San Francisco where the concentration of biotechnology firms is high (Deeds *et al.*, 1997). Although firms that have outstanding technology will attract high-quality researchers even if they are located elsewhere in the country, being centrally located enhances a firm's ability to compete, even for well-established firms. A dummy variable for *location* took a value of 1 if the main offices of a young biotechnology company were located in one of the following areas that were consistently rated among the top three biotechnology locations for the period of our

study (Burrill and Lee, 1990, 1993; Lee and Burrill, 1995): San Francisco, Boston, or San Diego. *Location* took on a value of 0 otherwise.

We also controlled for technological uncertainty associated with the *product stage* of the firm's lead product at the time of the IPO; we reviewed the company sections of the prospectuses to determine how advanced the firm's technology was (Pisano, 1991; Pisano and Mang, 1993). We coded the product that was at the latest stage into one of the following nine categories: discovery stage, research and development, preclinical indication, phases I through III clinical trials, new drug approval (NDA) filing/FDA approval pending, final market approval, and revenue-generating, relatively speaking. We also examined the use-of-proceeds section of the prospectus to confirm that the lead product, as defined above, was also that which was designated to receive the most funding.

In addition, we included a control variable for the *size of the upper echelon*, since prior research has found this to be positively related to the success of entrepreneurial firms (Eisenhardt and Schoonhoven, 1996). For this measure, we counted the number of managing officers and outside board members who were listed on the firm's final prospectus. We also included a variable for the *average prior position level* held by members of the firm's upper echelon, which reflects the caliber of the prior jobs the executives held. To assess the *average prior position level* of the over 3200 people executives in our database, we created a 0–5 ranking, ranging from low to high, beginning with nonmanagement positions and ending with CEO/president, similar to that employed by Eisenhardt and Schoonhoven (1996), and we used this scheme to code each of the members of our upper echelons. From these data, we calculated the mean level of prior position for each upper echelon in our dataset.

Finally, we included a variable that accounts for the type of business the biotechnology company was in. Biotechnology is a fairly segmented industry. The applications of its products range from tissue repair in humans to agricultural products. From the main company description in the prospectuses, firms were coded as being in therapeutics, diagnostics, both diagnostics and therapeutics, agriculture, chemical, or other. To verify the firm's business, we referred to the IBI database and BioScan. For the private companies, we had information on up

to 30 such categories. For this *business type* variable, we coded whether the company was in a core biotechnology field (i.e., therapeutics or therapeutics and diagnostics, or not).

Analysis

We used Heckman selection models to guard against the possibility of sample selection bias (Heckman, 1979). In general, sample selection can arise when the criteria for selecting observations are not independent of the outcome variables. Here, since we are studying factors that influence financial indicators of IPO success, which can only be measured when a firm goes public, we want to guard against the possibility that there is some other factor, in addition to those we study, that accounts for the likelihood of firms being able to go public in the first instance.

Heckman's procedure generates consistent, asymptotically efficient estimates that can enable us to generalize to the larger population of biotechnology firms (cf. Heckman, 1979). In essence, the Heckman model is a two-stage procedure that uses the larger risk set of public and private firms, including firms that ceased to exist as of 1996 in both categories ($n = 858$). Probit regression was used to estimate the likelihood of completing an IPO during the first stage, and estimates of parameters from that model were then incorporated into a second-stage regression model to predict IPO success (Van de Ven and van Praag, 1981). For the first-stage model, we used the information we had available for our public and private firms—geographical location, year of founding, and type of business—to predict likelihood of going public.² In the second stage, although the sample includes all 858 firms, the standard errors reported reflect the smaller sample of firms ($n = 299$).

To account for the fact that we had financial information that spanned two decades, we transformed our IPO success estimates into constant 1996 dollars and logged the estimates for our firms. Finally, for all of our analyses, we included the *equity index* variable described earlier, which

² We note that two-stage models do a particularly good job at estimation when there is at least one variable that may be considered an 'instrument' that is a good predictor in the first stage but not the second stage of the model; in this case, that 'instrument' was business type (see Winship and Mare, 1992, for further discussion).

provides an indicator of the environmental conditions the firm faced when trying to go public. The numbers we used were calibrated not just by the year but also by the month preceding the offering, which produces fairly fine-grained estimates.

RESULTS

Correlations between the main variables in this study are provided in Table 1. Table 2 presents the results from Heckman selection models in which the first stage predicted whether or not a company was able to go public and in which the second stage predicted IPO success. The first-stage probit models predicting whether a company was able to go public correctly classified 73 percent of the cases. As shown in Table 2, the analyses predicting IPO success begin with models that include firm and industry-level control variables, then include the main effects for firm partnerships and equity market uncertainty and then include the interaction terms between equity market uncertainty and the specific forms of endorsement relations and strategic alliances, as suggested in Hypotheses 1–3.

Hypothesis 1 predicted that having a prestigious VC partner when a firm goes public would be particularly beneficial to IPO success when the equity markets are cold for new issues. Model III in Table 2 tests this hypothesis. Since we operationalized equity market uncertainty as a continuous measure ranging from cold to hot, then support for Hypothesis 1 would be indicated if our results showed a significant and negative interaction effect between the *equity market index* and *VC prominence*. Our findings reveal a significant and negative interaction, which supports Hypothesis 1.

The results also support Hypothesis 2. Hypothesis 2 was that underwriter prestige would be positively related to IPO success, particularly when the equity markets are relatively hot for new issues. As shown in Model IV of Table 2, the interaction term between *equity index* and *underwriter prestige* was positive and significant, as predicted. The main effect for underwriter prestige also remained significant and positively related to IPO success in all of our models, consistent with prior research (e.g., Carter and Manaster, 1990). In Model V of Table 2, we included all of the interaction terms. The results supporting Hypotheses 1 and 2 remained significant and in the directions predicted.

Hypothesis 3 predicted that prominent downstream strategic alliances would be positively related to IPO success, especially when the equity markets are relatively cold for new issues. This hypothesis was not supported; as shown in Model III in Table 2, the interaction effect between equity market uncertainty and downstream alliances was not significant. In addition, we found no main effects for strategic alliances on IPO success.

With respect to our control variables, firm size and the amount of private financing a firm received prior to the IPO were significant, as was the size of a firm's upper echelon. And, as shown in two of the baseline models in our tables, firms with more advanced products tended to have more successful IPOs. With respect to the main effects, as expected, having a prestigious underwriter on board was consistently and positively related to IPO success. The main effect for VC prominence was marginally significant and positively related to IPO success as well. These latter results, when considered in tandem with the effects found for our interaction terms, provide strong evidence for the importance of considering both what types and when interorganizational ties benefit firms.

DISCUSSION

The present study demonstrates that various types of interorganizational ties have differential effects on IPO performance and this, in turn, is contingent on the favorability of the equity markets. Results from this sample of biotechnology IPO firms show that young firms benefit from having partners with prominent organizations in different ways and at different times: whereas partnerships with prestigious VCs positively affect IPO success when the equity markets are relatively cold for new issues, partnerships with prestigious underwriters positively affect IPO success when the equity markets are relatively hot for new issues. Our results did not, however, yield significant effects for the contingent value of a new venture's strategic alliance partnerships on IPO success. The findings confirm our general thesis that network effects are not uniform but rather are contingent upon both the nature of a firm's ties and upon the uncertainty associated with the marketplace.

When investigating the contingent value of interorganizational embeddedness, we focused on two

Table 1. Means, standard deviations, and correlations (*N* = 299)

Variable	<i>X</i>	S.D.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Firm age	4.90	2.89														
2. Firm size	85.64	120.85	0.08													
3. Location	0.50	0.50	0.07	0.05												
4. Product stage	4.65	2.82	0.23***	0.27***	0.07											
5. Private financing ^a	6.92	0.77	0.10†	0.28***	0.19***	0.12*										
6. Size of upper echelon	10.86	3.47	0.12*	0.29***	0.20***	0.11*	0.51***									
7. Avg. prior position of upper echelon	2.72	0.67	-0.08	-0.08	0.01	0.11†	0.06	-0.06								
8. Equity index	3.78	0.96	0.08	0.02	0.15**	0.02	0.19***	0.13*	-0.00							
9. VC prominence	0.32	0.47	-0.01	0.05	0.33***	0.04	0.20***	0.20***	0.05	0.17**						
10. Underwriter prestige ^b	7.63	1.95	0.13*	0.22***	0.12†	0.17**	0.41***	0.42***	0.13*	0.14*	0.27***					
11. Downstream alliances	0.52	0.88	0.09	0.05	0.05	-0.17**	0.24***	0.27***	-0.02	0.06	0.13*	0.20**				
12. Equity index × VC	0.07	0.43	0.02	0.01	-0.02	0.01	-0.07	-0.05	0.01	-0.07	0.14*	-0.05	-0.02			
13. Equity index × prominence underwriter prestige ^b	0.25	1.99	0.10	-0.07	-0.13*	-0.08	-0.13*	-0.13*	-0.13*	-0.09	-0.05	-0.27***	-0.02	0.26***		
14. Equity index × downstream alliances	0.05	0.82	-0.03	0.03	0.10	-0.06	-0.04	0.02	0.11†	-0.04	-0.02	-0.01	0.06	0.12*	0.20**	
15. IPO success ^c	-0.01	0.87	0.18**	0.38***	0.25***	0.19***	0.58***	0.50***	0.04	0.17**	0.28***	0.54***	0.25***	-0.10†	-0.01	0.01

*** *p* ≤ 0.001; ** *p* ≤ 0.01; * *p* ≤ 0.05; † *p* ≤ 0.10
^a Adjusted to constant 1996 dollars, then logged.
^b *N* = 244 due to investment banks not ranked on Carter–Manaster scale.
^c Based upon average standardized scores for firm net proceeds, pre-money market value, 90-day market value, and 180-day market value, adjusted to constant 1996 dollars and logged.

Table 2. The effects of interorganizational partnerships on IPO success^{a,c}

	I	II	III	IV	V
<i>Control variables</i>					
Firm age	0.02 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)	-0.01 (0.01)
Firm size	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)	0.00*** (0.00)
Location	0.08 (0.09)	0.01 (0.09)	0.06 (0.09)	0.00 (0.10)	0.06 (0.09)
Product stage	0.03† (0.01)	0.02 (0.02)	0.02† (0.01)	0.03 (0.02)	0.02 (0.01)
Private financing ^b	0.39*** (0.06)	0.28*** (0.06)	0.27*** (0.06)	0.06*** (0.06)	0.27*** (0.06)
Size of upper echelon	0.06*** (0.01)	0.04** (0.01)	0.04** (0.01)	0.04* (0.01)	0.04** (0.01)
Avg. prior position of upper echelon	0.07 (0.06)	0.04 (0.06)	0.06 (0.06)	0.03 (0.06)	0.06 (0.06)
<i>Main effects</i>					
Equity index		0.05 (0.04)	0.05 (0.04)	0.05 (0.04)	0.05 (0.04)
VC prominence		0.14 (0.09)	0.16† (0.09)	0.14 (0.09)	0.16† (0.09)
Underwriter prestige		0.12*** (0.02)	0.14*** (0.02)	0.12*** (0.02)	0.14*** (0.02)
Strategic alliances		0.05 (0.05)	0.05 (0.04)	0.05 (0.05)	0.05 (0.04)
<i>Interaction effects</i>					
Endorsement relations					
Equity index × VC prominence			-0.28** (0.09)		-0.28** (0.09)
Equity index × underwriter prestige			0.09*** (0.02)		0.09*** (0.02)
Strategic alliances					
Equity index × downstream alliances				0.03 (0.05)	-0.01 (0.04)
Constant	-03.03*** (0.43)	-1.93*** (0.49)	-2.00*** (0.47)	-1.94*** (0.49)	-1.99*** (0.47)
Wald chi-square	275.58***	272.08***	319.75***	272.70***	319.81***
Rho	-0.49	-0.36	-0.31	-0.36	-0.31
N	299	244	244	244	244

*** $p < 0.001$ (two-tailed tests); ** $p < 0.01$; * $p < 0.05$; † $p < 0.10$ ^a Unstandardized regression coefficients reported; standard errors in parentheses.^b Adjusted to 1996 dollars and logged.^c Based upon average standardized scores for firm net proceeds, pre-money market value, 90-day market value, and 180-day market value, adjusted to 1996 dollars and logged.

related questions: which ties matter (network factors) and when ties matter (contextual factors). As depicted in Fig. 1, the network factor was broken down into three types of ties: VC partnerships, investment bank partnerships, and strategic alliances. Market uncertainty was conceptualized here as whether the equity markets were 'hot' or 'cold' for new issues. Together, these two dimensions formed the basis of our investigation that extends our understanding of the contingent value of interorganizational endorsements and alliances.

The present research extends prior research on interorganizational networks and entrepreneurship in several respects. First, it expands upon the growing research in strategy and organizations on social networks. Most of the prior research on interorganizational ties has focused on the main effects of having ties to prominent actors (e.g., Granovetter, 1985; Gulati, 1998; Mizruchi and Stearns, 1994). Often, this work has presented the embeddedness perspective as a more social account of strategic behavior of organizations than traditional economic theories of firm behavior (see, for example, Uzzi, 1999). Building on this influential stream of organizational and strategic management research on embeddedness, we investigated the conditional effects of interorganizational embeddedness. While a contingency perspective on embeddedness has begun to be advanced for individual actors in pursuit of individual goals (e.g., Podolny and Baron, 1997), rarely have such boundary considerations been explored for firms and their pursuit of organizational goals.

To develop our contingency perspective, we built upon theories of the social organization of attention and decision-making (Simon, 1947/1997; Ocasio, 1997). Whereas prior research has suggested that certain prevailing industry-based logics may focus the attention of a firm's executives (Ocasio, 1997; Thornton and Ocasio, 1999), our work considers how external parties', such as investors', attention can be altered by the market context and how this, in turn, can affect firm performance. We argue that during cold markets investors' concerns center on avoiding Type I errors—that is, missing out on worthwhile investments—whereas during hot markets investors' concerns center on avoiding Type II errors—that is, investing in firms that are not worthwhile. Our theory suggests that investors resolve these concerns by attending to the signals they receive from the endorsements and partnerships of the focal

firm. Future research that examines further how the attention of key external parties is shaped by the market context which surrounds a focal firm would extend the present research.

Future research on the contingent value of interorganizational relationships could also extend the present study by considering the actual content of information that flows across different network ties (e.g., Westphal, Gulati, and Shortell, 1997; Gulati and Westphal, 1999). As scholars have suggested, network studies tend to focus on the structure of relationships between actors at the expense of considering the content of information, resources, and/or access that flows between ties (Podolny and Baron, 1997; Uzzi, 1996; Hansen, 1999). In the IPO context, future research could investigate the specific forms of joint venture agreements, licensing agreements, and/or strategic alliances in which firms engage to determine whether differences in the nature of the resources flowing between a young firm and its partners affect investor perceptions and, hence, IPO success.

Additionally, scholars could examine the degree of closeness between endorsement and partner firms to a focal firm and their effects on outsider perceptions: for example, it could be that the duration of the VC relationship matters more to institutional investors than VC prestige under different market conditions, affecting IPO success. Scholars could also examine the effects that different network clusters of intermediaries have on IPO success: having different investment bank syndicate structures may send different types of signals to institutional investors, affecting their perceptions of a new venture's potential under varying market conditions.

In addition to contributing to research on social networks, the present study extends prior research on the sources and implications of uncertainty for new venture performance. Prior research has conceptualized uncertainty for startup firms as associated with firm characteristics such as firm age and location (e.g., Sorenson and Audia, 2000). We extend these studies of endogenous sources of uncertainty by investigating the effects of one exogenous source of uncertainty: uncertainty associated with the equity markets. In addition, by considering how the nature of the concerns that investors face during a hot equity market differ substantially from the nature of the concerns that investors face during a cold equity market, we

broaden previous unidimensional conceptualizations of uncertainty which has tended to characterize uncertainty as either 'high' or 'low' (e.g., Stuart *et al.*, 1999). Our perspective is consistent with the view that uncertainty is a multidimensional phenomenon (Milliken, 1987) which can alter the prevailing logic of strategic decision-making.

The present research extends prior research on the implications of uncertainty for firm performance as well. Prior studies have focused on the effects of uncertainty on firm strategic decision-making such as the scope of a firm resulting from decisions regarding vertical integration (Sutcliffe and Zaheer, 1998). In the present study, we considered how the uncertainty impacting third-party stakeholders affects the performance of young firms. Thus, this study examines the implications of uncertainty for interorganizational perceptions which, in turn, affects firm performance. In mediated markets, such as the IPO, in which the behavior of multiple stakeholders can affect the performance of firms, such an expanded purview may be useful.

Future research may consider other contextual factors as well as equity market uncertainty when exploring the benefits to young firms of having ties with prominent others. Scholars could consider market indicators such as interest rate volatility (e.g., Bae *et al.*, 1999) or regulatory changes that might affect the probability of drugs being successful on the market, or even interindustry effects as investors switch out of and into different markets due to new product innovation or scientific advancement. There may be differences worth exploring that may be more directly tied to investor considerations than the industry-specific measure of market factors employed in the present study.

While considering possible opportunities to expand upon the present research, two issues are worth considering. First, the present study investigates firms during the period 1979–96 and so may not capture recent changes in the roles and preferences of financial institutions and their evaluation and endorsement of young firms. One example of such changes concerns the roles that financial institutions play in the financial markets; recent work on the changing activities of banks (Tkacs, 1998) has suggested that investment banks have recently moved into the private equity market. Thus, investment banks and investment banking arms of commercial banks are reported to have begun to engage in due diligence activities

that very much resemble the early-stage review processes that VCs typically engage in. Such activities dilute the contrast set forth in the present research between investment banks' choosing of young firms at the time of IPO and VCs' taking of firms public. Future research that examines the nature of and changes in investment bank and VC deal flow and organizational structures over the past decade would lend insight into if and how our findings can be generalized upon.

Our study is also limited by its focus on a single industry. Although the biotechnology industry is an industry in which the needs for cash are high and so the IPO event is a particularly worthwhile context for studying the effects of organizational embeddedness on a firm's ability to obtain financial capital, it is also the case that this is an industry based on new technology in which the uncertainty associated with firm evaluation is especially strong (Aldrich and Fiol, 1994). As described, firms generally go public after approximately 4 or 5 years of existence, yet the product development life cycle of biotechnology products typically lasts between 7 and 10 years (Burrill and Lee, 1993). At the time of IPO, then, investors evaluate biotechnology firms which have yet to internally generate their own revenues. Thus, we have chosen a strong situation in which to test our hypotheses.

The results of this study do open up several avenues for future research that push further the contingency perspectives set forth in prior organizational and strategic management research. While there is a growing body of research across the strategy field seeking to identify contingencies and boundary conditions for important theoretical claims, this lens was yet to be applied to the burgeoning literature on social embeddedness and strategic behavior and outcomes. We deviate here from the general claim that more ties are better and provide a richer account of the relative importance of various kinds of ties and of how their importance may also change with the market context. As our results show, partnerships with prominent firms are not homogeneous in their effects for young firms but rather depend upon the types of ties as well as the market context in which such investment decisions are made. Together, these two contingency dimensions can enhance our understanding of the conditions under which interorganizational partnerships affect the success of young firms.

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